

LESSON PLAN	COURSE: MDM4U
Date:	Unit of Study: Unit 2
Materials/Resources: Mathmetics of data management study guide	Lesson Focus/Theme: Chapter 4: Permutations and organized counting
Prior Knowledge: permutation	
Overall Expectations: By the end of this lesson, students will: solve problems involving the application of permutations and combinations to determine the probability of an event.	
Specific Expectations: 2.3 solve introductory counting problems involving the additive counting principle (e.g., determining the number of ways of selecting 2 boys or 2 girls from a group of 4 boys and 5 girls) and the multiplicative counting principle (e.g., determining the number of ways of selecting 2 boys and 2 girls from a group of 4 boys and 5 girls) 2.4 make connections, through investigation, between combinations (i.e., $n \text{ choose } r$) and Pascal's triangle [e.g., between $(2 \text{ } r)$ and row 3 of Pascal's triangle, between $(n \text{ } 2)$ and diagonal 3 of Pascal's triangle] 2.5 solve probability problems using counting principles for situations involving equally likely outcomes	
Learning Goals: By the end of this lesson, students will know permutation with identical element and pascal's triangle	

Success Criteria:

By the end of this lesson, I will be able to do permutations with some identical elements

By the end of this lesson, I will be able to use Pascal's triangle

By the end of this lesson, I will be able to apply pascal's triangle method

Assessment Strategies: Tests, quiz and project**Time:**
10 min**Introduction/Minds-On:**

Answer questions students don't understand from last class if any

Time:
100 min**Lesson:**

Chapter 4: Permutations and organized counting

4.3 Permutations with some identical elements

4.4 Applying Pascal's method

Time:

10 min

Extension / Wrap-up:

Review the key concepts and formula for this lesson

Notes / Reminders / Homework:

Review for quiz next class

Lesson Reflection:

